

Experiment No: 1

Experiment Name: Find the Bigger Number

Task Name

Write a Java program to determine the bigger number between two numbers.

Objective

- To learn how to take input from the user
- To understand conditional statements in Java
- To compare two numbers

Problem Analysis

In this program, the user enters two numbers and the program determines which number is bigger using an if-else condition.

Input: Two numbers from the user

Process: Compare the numbers using if-else condition

Output: Display the bigger number

Algorithm

1. Start
2. Import Scanner class
3. Take two numbers as input
4. Compare numbers using if condition
5. Display the bigger number
6. End

Source Code:

```
BiggerNumber.java > BiggerNumber
1  import java.util.Scanner;
2
3  public class BiggerNumber {
4      public static void main(String[] args) {
5
6          Scanner input = new Scanner(System.in);
7
8          System.out.print("Enter first number: ");
9          int num1 = input.nextInt();
10
11         System.out.print("Enter second number: ");
12         int num2 = input.nextInt();
13
14         if(num1 > num2)
15         {
16             System.out.println("Bigger number is: " + num1);
17         }
18         else
19         {
20             System.out.println("Bigger number is: " + num2);
21         }
22     }
23 }
24
25
```

Output:

```
.java\jdt_ws\Lab report-2_fb5da8d5\bin' 'Bigge
Enter first number: 12
Enter second number: 13
Bigger number is: 13
PS C:\Users\dell\Downloads\OneDrive\Documents\
t-2> █
```

Discussion

In this experiment, I learned how to take input using Scanner and compare two numbers using an if-else condition.

Experiment No: 2

Experiment Name

Simple Calculator Using Switch Statement

Task Name

Write a Java program that asks the user for two numbers and an operator (+, -, *, /) and performs the selected operation.

Objective

- To learn how to use switch statements in Java
- To perform arithmetic operations
- To understand user input handling

Problem Analysis

In this program, the user enters two numbers and an operator. The program checks the operator and performs the corresponding arithmetic operation using a switch statement.

Input: Two numbers and an operator from the user

Process: Perform arithmetic operation using switch statement

Output: Display the calculated result

Algorithm

1. Start
2. Import Scanner class

3. Take first number as input
4. Take second number as input
5. Take operator input
6. Use switch statement to perform calculation
7. Display the result
8. End

Source Code:

```
1  import java.util.Scanner;
2  public class CalculatorSwitch {
3      public static void main(String[] args) {
4          Scanner input = new Scanner(System.in);
5
6          System.out.print("Enter first number: ");
7          double num1 = input.nextDouble();
8
9          System.out.print("Enter second number: ");
10         double num2 = input.nextDouble();
11
12         System.out.print("Enter operator (+, -, *, /): ");
13         char operator = input.next().charAt(0);
14
15         switch(operator)
16         {
17             case '+':
18                 System.out.println("Result = " + (num1 + num2));
19                 break;
20
21             case '-':
22                 System.out.println("Result = " + (num1 - num2));
23                 break;
24
25             case '*':
26                 System.out.println("Result = " + (num1 * num2));
27                 break;
28
29             case '/':
30                 System.out.println("Result = " + (num1 / num2));
31                 break;
32
33             default:
34                 System.out.println("Invalid Operator");
35         }
36     }
37 }
```

Output:

```
Enter first number: 12
Enter second number: 10
Enter operator (+, -, *, /): +
Result = 22.0
PS C:\Users\dell\Downloads\OneDr
```

Discussion

In this experiment, I learned how to use switch statements in Java to perform different arithmetic operations based on user input. The program takes two numbers and an operator from the user and calculates the result accordingly.